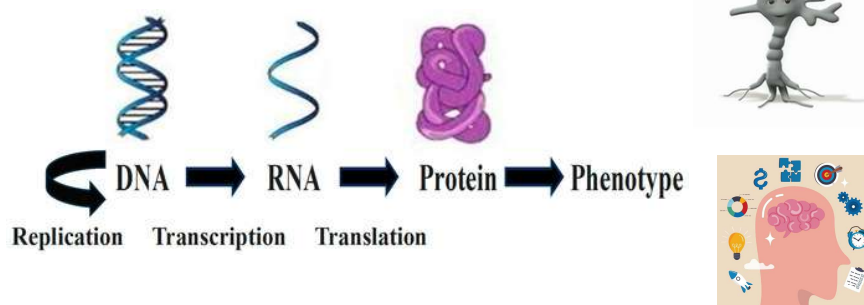


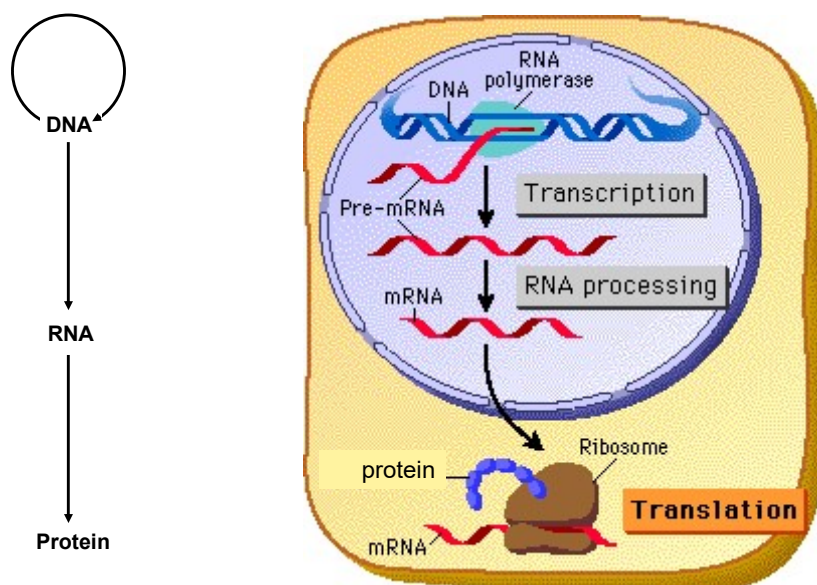
Basic molecular biology and Protein science

鄭茵若

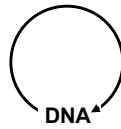


The central dogma

DNA Encodes RNA; RNA Encodes Protein.



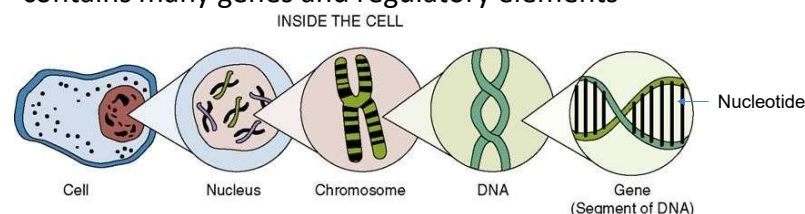
The big three molecules for genetics



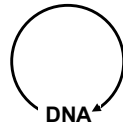
- **DNA (DeoxyriboNucleic Acid)** 
 - Heredity genetic information of most of the organisms
 - Store in nucleus
- **RNA (RiboNucleic Acid)** 
 - Bring short pieces of information to the cytoplasm
 - Provide templates to synthesize protein
- **Protein** 
 - Each protein has one or more specific functions
 - They control the structure and function of the cells

The building structure of life

- **Gene**
 - basic physical and functional units of heredity.
 - A specific sequence of DNA encodes instructions on how to make a protein.
- **Genome**
 - an organism's complete set of genetic information.
- **Chromosome**
 - an organized structure of DNA and protein in the nucleus.
 - contains many genes and regulatory elements



The central dogma



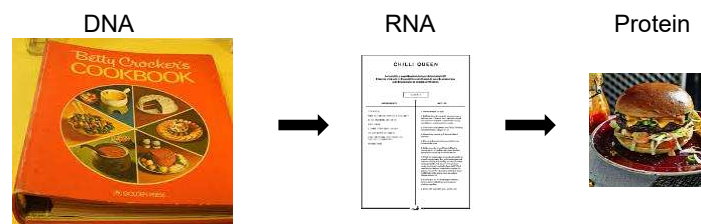
DNA

RNA

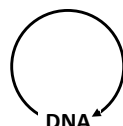
Protein

•Proposed by Francis Crick in 1958 to describe the flow of information in a cell.

•Information stored in DNA is transferred residue-by-residue to RNA, which in turn transfers the information residue-by-residue to protein.



The central dogma



DNA

RNA

Protein

Replication

Duplication of DNA using DNA as the template

Require: **DNA polymerase**

polymer 聚合物
-ase 酶 (酵素)

Transcription

Synthesis of RNA using DNA as the template

Require: **RNA polymerase**

Translation

Synthesis of proteins using RNA as the template

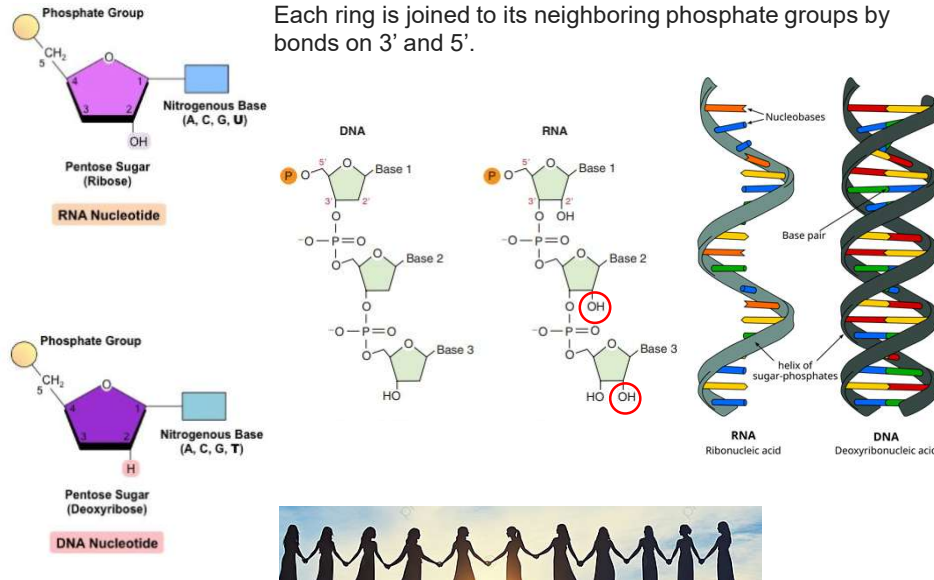
Require: **Ribosome**

-some = a group of
聚合體

Nucleotides-The building block of DNA or RNA

Sugar ring contains five carbon

Each ring is joined to its neighboring phosphate groups by bonds on 3' and 5'.

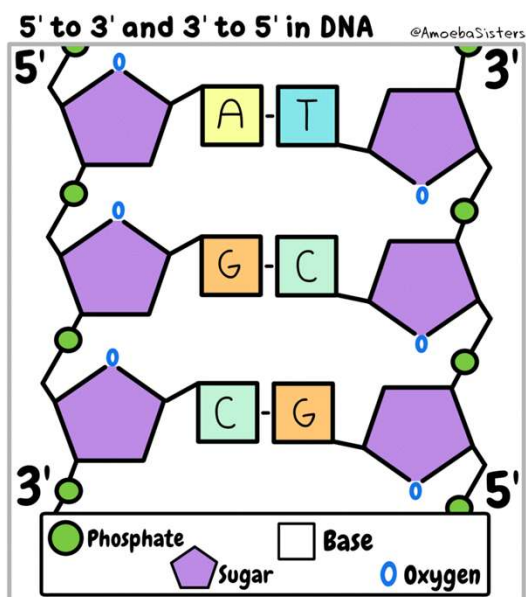


DNA & RNA direction

Both DNA and RNA are synthesized only from 5' → 3'

DNA or RNA sequence is "read" from 5' to 3'

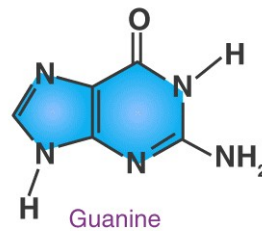
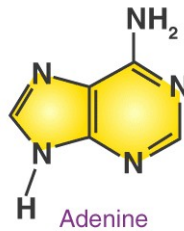
DNA strands are antiparallel. One strand runs in the 5' → 3', while the **complementary strand** runs in the 3' → 5'



The bases for DNA and RNA

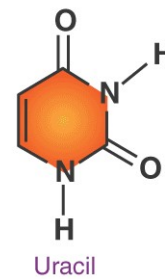
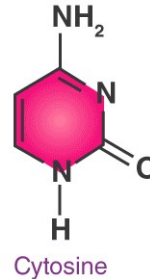
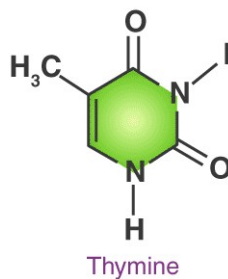
DNA

Adenine (A)
Cytosine (C)
Guanine (G)
Thymine (T)



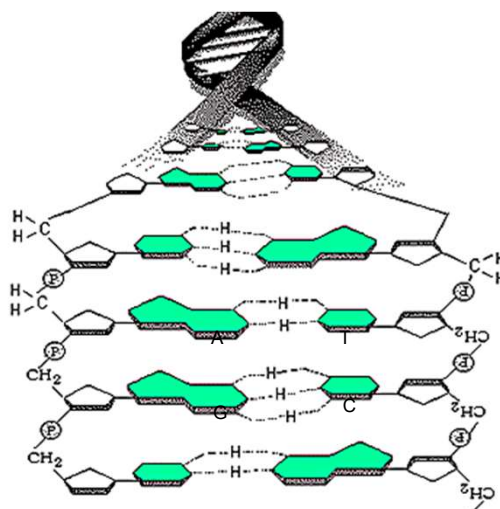
RNA

Adenine (A)
Cytosine (C)
Guanine (G)
Uracil (U)

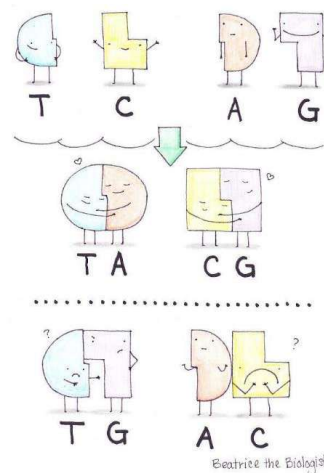


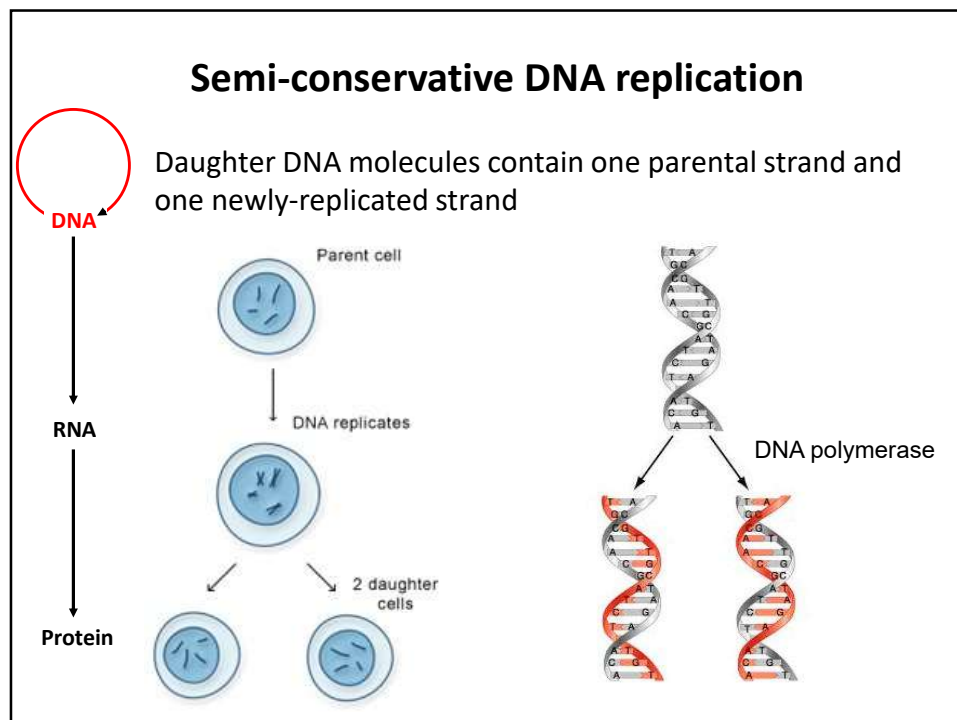
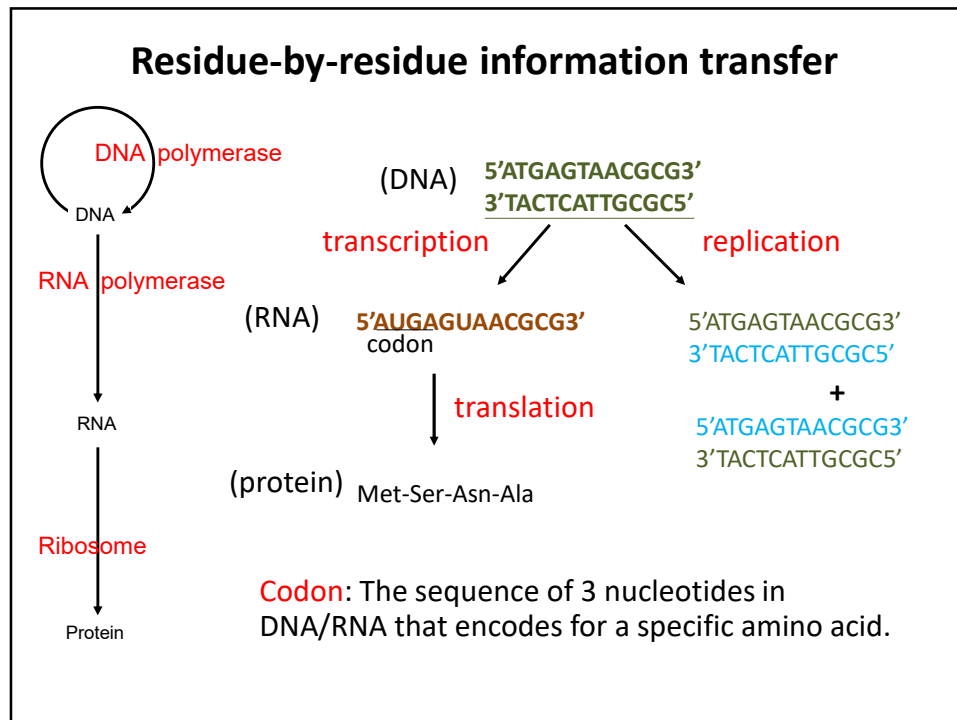
Base pairing

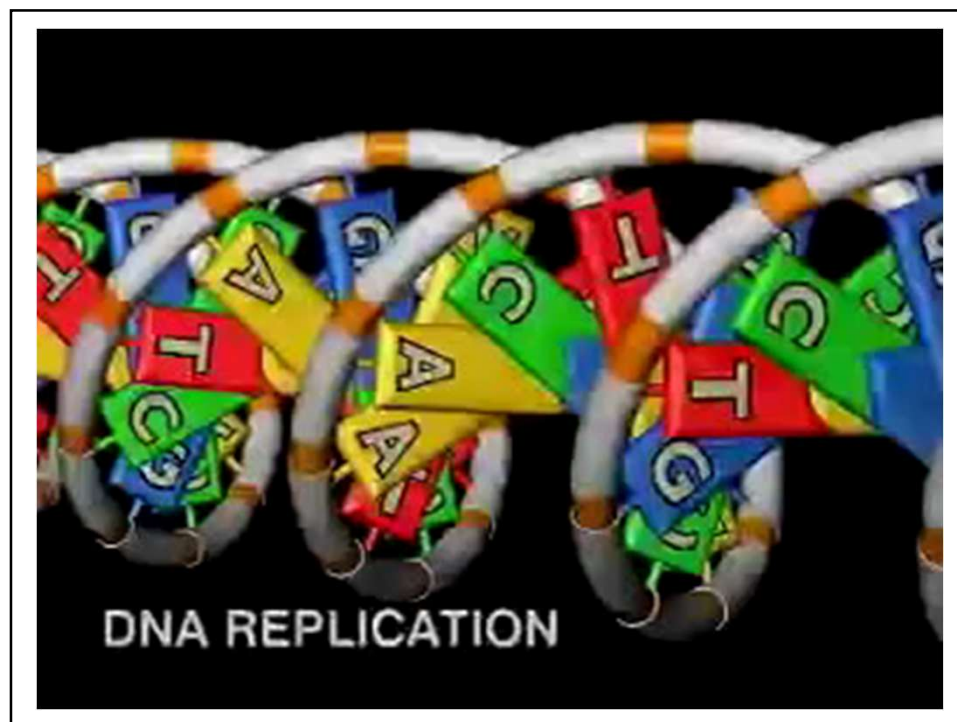
A base pair is a unit of two complementary nucleotide bases connected by hydrogen bonds that form the DNA or RNA double helix



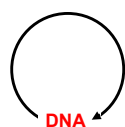
Rule: A-T(U) and G-C







Transcription



- Making messenger RNA (mRNA) based on information from part of DNA by **RNA polymerase**

- Rule for base pairing:

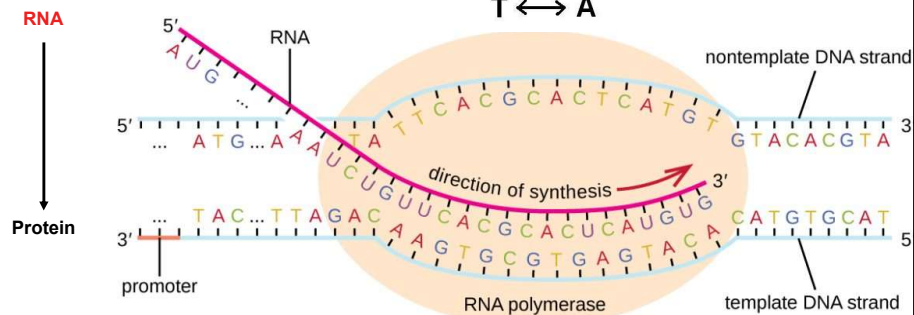
DNA RNA

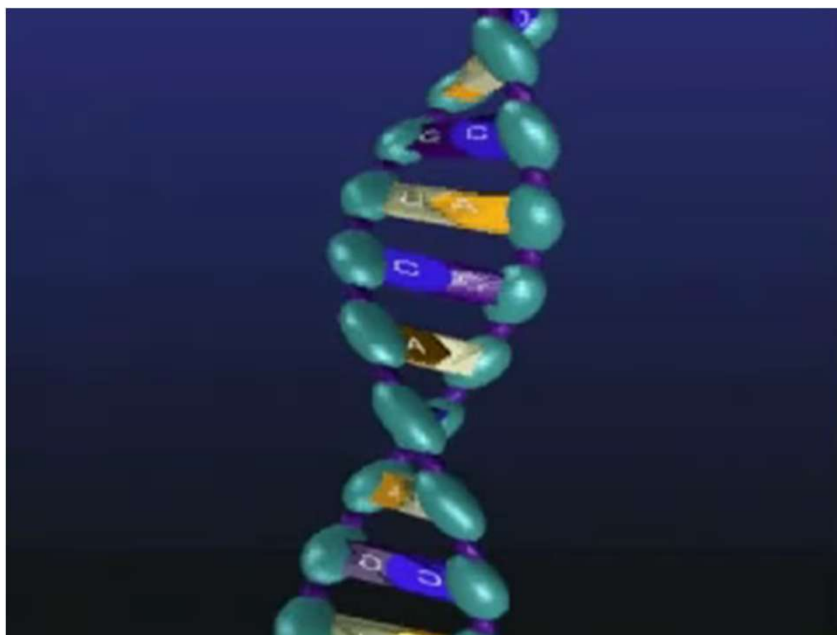
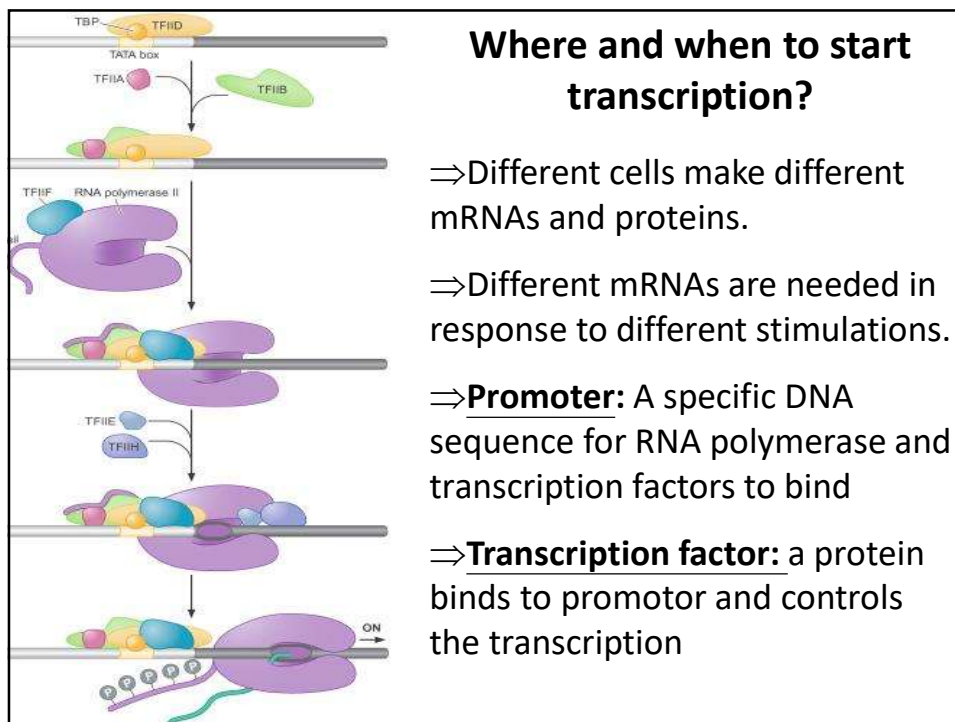
A $\leftrightarrow$ U

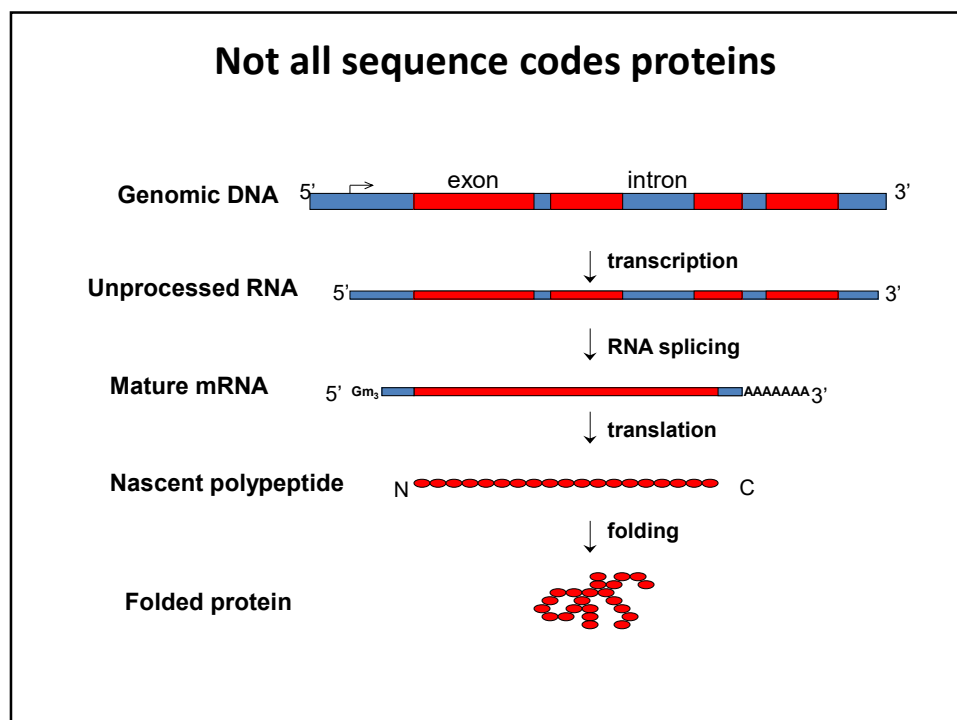
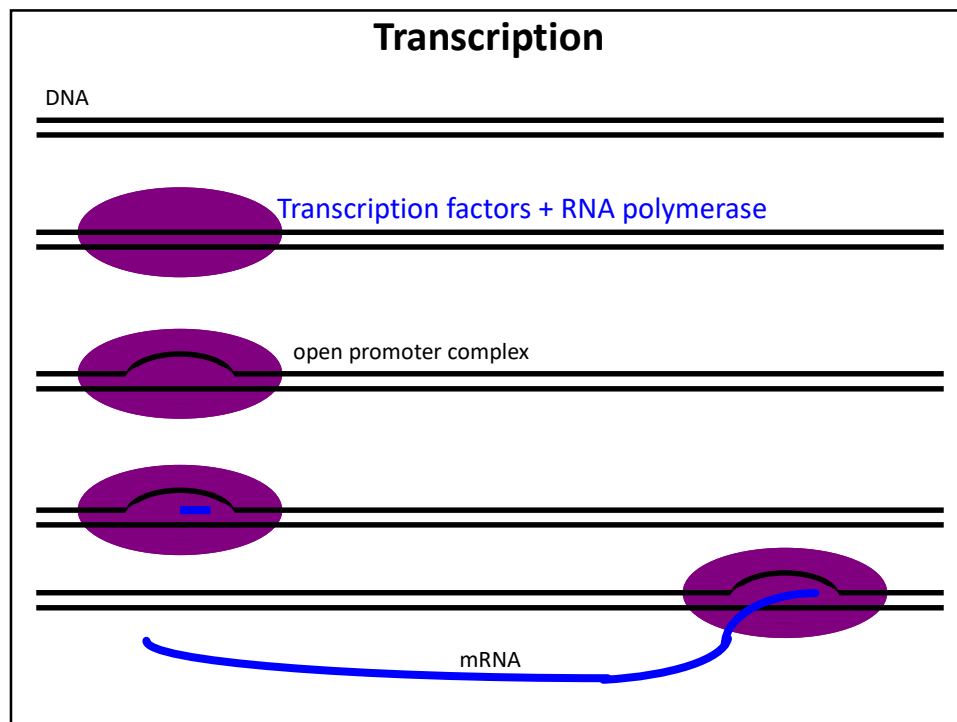
G $\leftrightarrow$ C

C $\leftrightarrow$ G

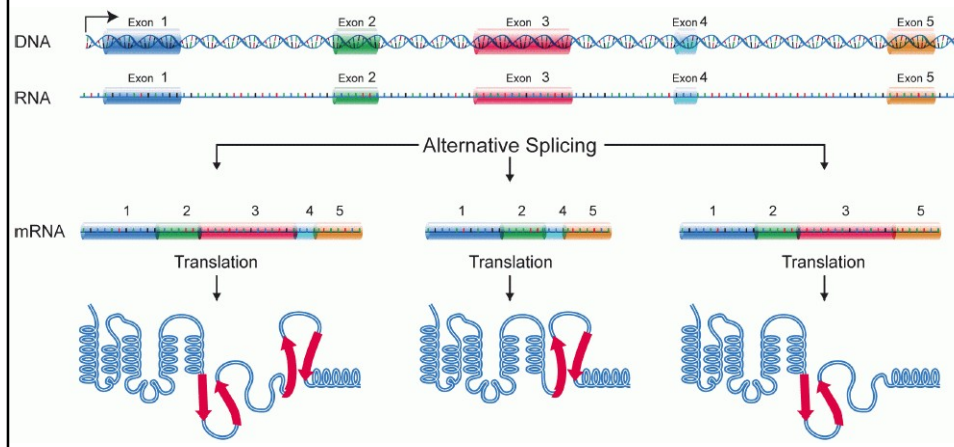
T $\leftrightarrow$ A



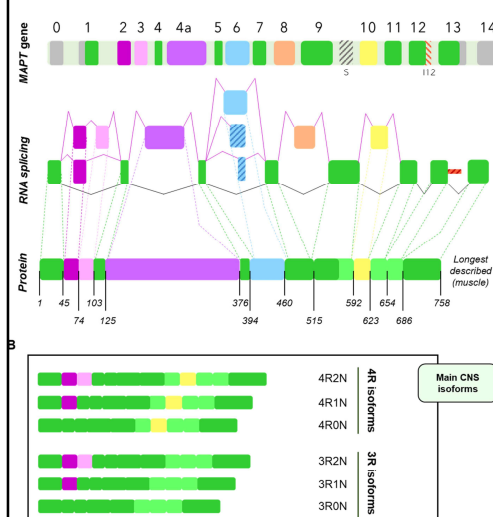




- **Exons:** Nucleotide sequences code for proteins.
- **Introns:** sequences that are spliced out during mRNA maturation and do not code for proteins.
- **Alternative Splicing:** different combinations of exons are joined together after introns are removed.



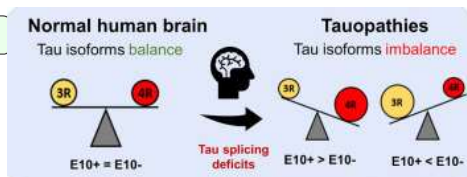
Alternative splicing to generate Tau isoforms



Tau isoforms are differentially sorted across neuronal compartments and have distinct functions.

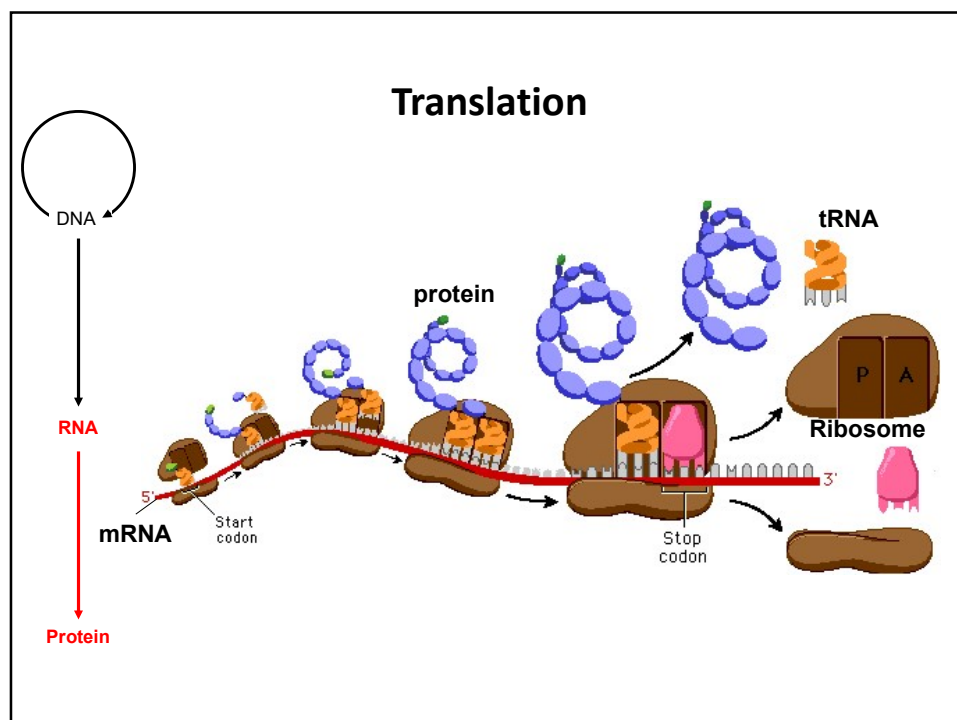
4R-Tau: High Stability & Fast Transport
3R-Tau: Structural Flexibility & Plasticity

0N, 1N, 2N: exist in different neuron parts, developmental stages, and



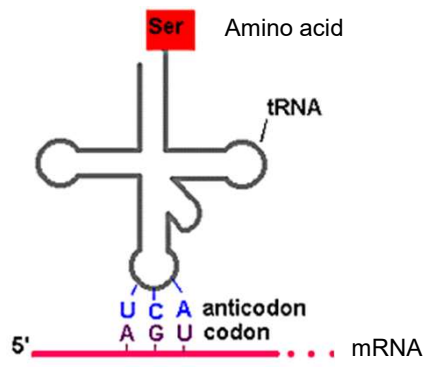
Types of RNA

- **Messenger RNA (mRNA)** carries information from DNA to the ribosomes for protein synthesis.
- **Non-coding RNA** is RNA that is not translated into a protein. Such as: tRNA, rRNA, shRNA, microRNA, siRNA....
 - **Transfer RNA (tRNA)** transfers a specific amino acid to the ribosome during translation.
 - **MicroRNA (miRNA)** functions in RNA silencing and post-transcriptional regulation of gene expression.
 - **Ribosomal RNA (rRNA)** is a component of the ribosomes, and makes up at least 80% of the RNA molecules in a cell

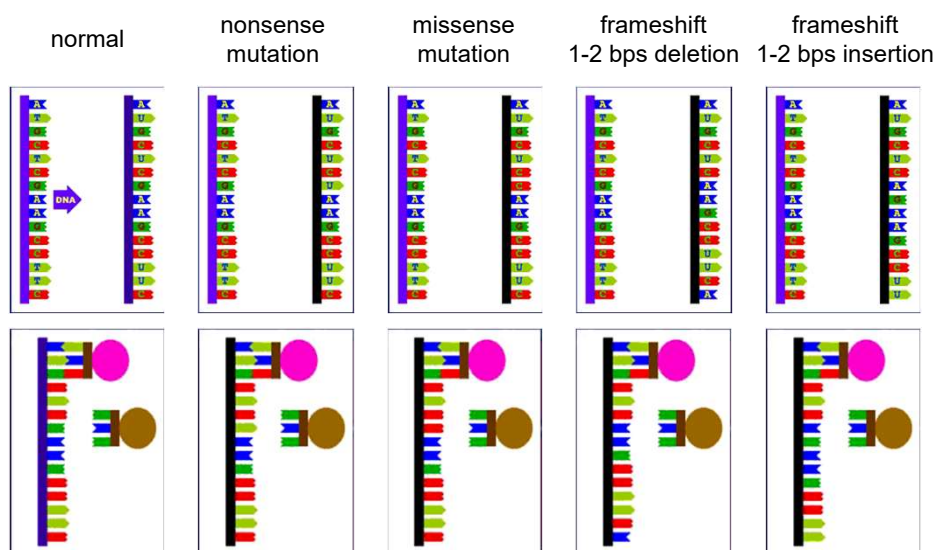


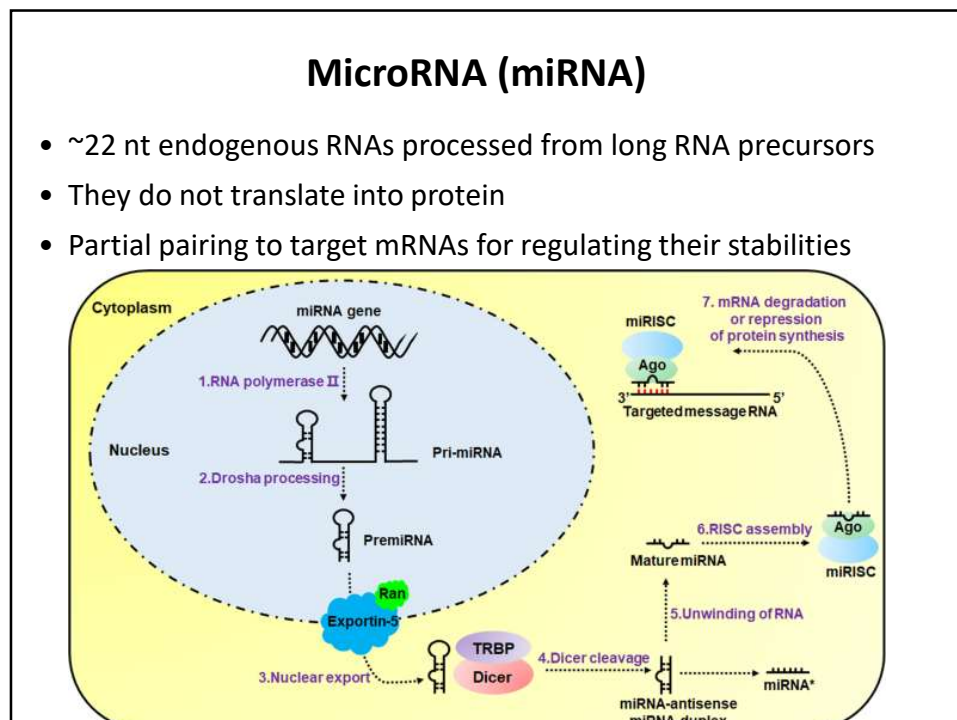
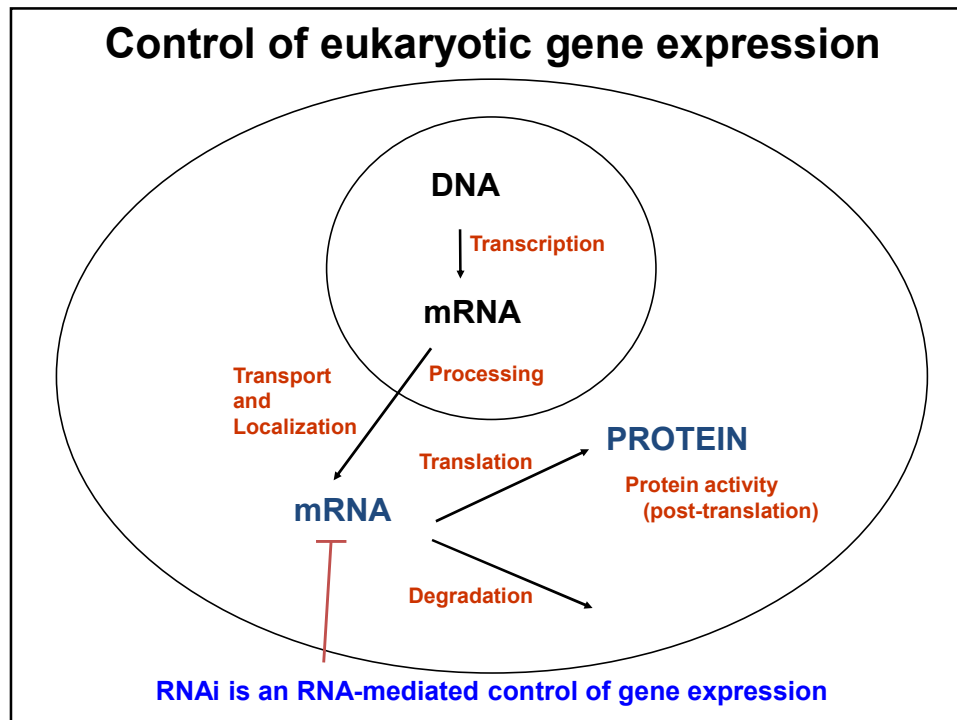
tRNA (Transfer RNA)

- A small RNA (~74-95 nucleotides) that transfers a specific amino acid to ribosome during translation.
- Each tRNA contains a 3 nucleotide sequence called anticodon, that can base-pair with its complementary codon in the mRNA.



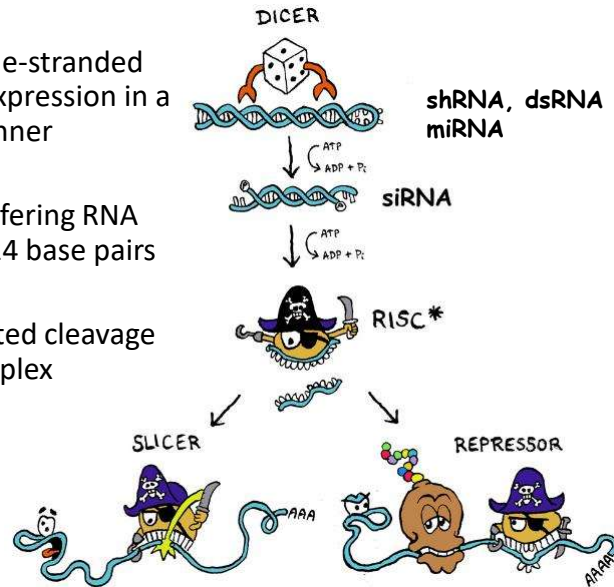
Types of DNA Mutations





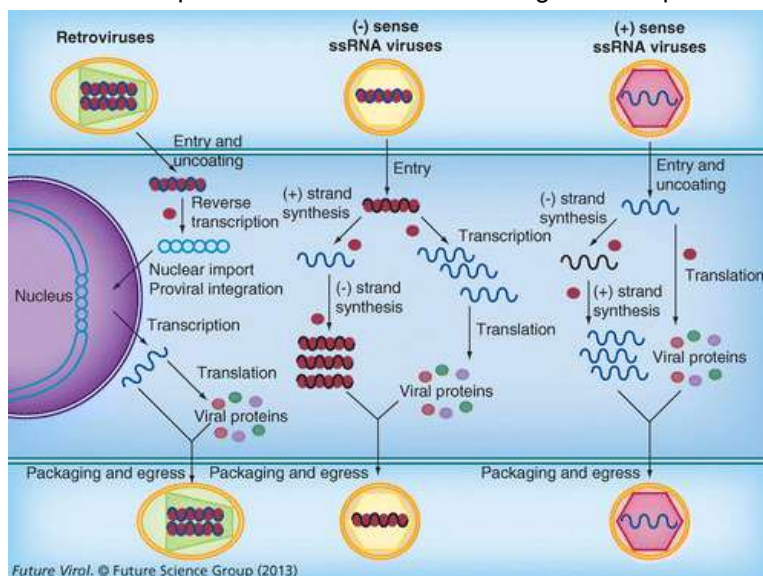
RNA interference (RNAi): post-transcriptional gene silencing

- RNAi – ability of double-stranded RNA to silence gene expression in a sequence-specific manner
- Induced by small interfering RNA (siRNA), typically 20–24 base pairs
- Silencing: siRNA-directed cleavage of mRNA via RISC complex

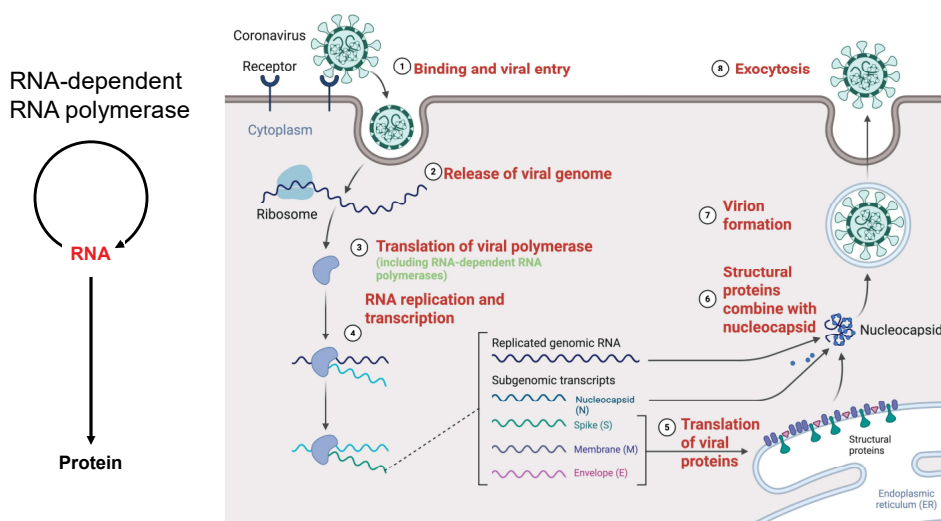


RNA as genetic material (RNA virus)

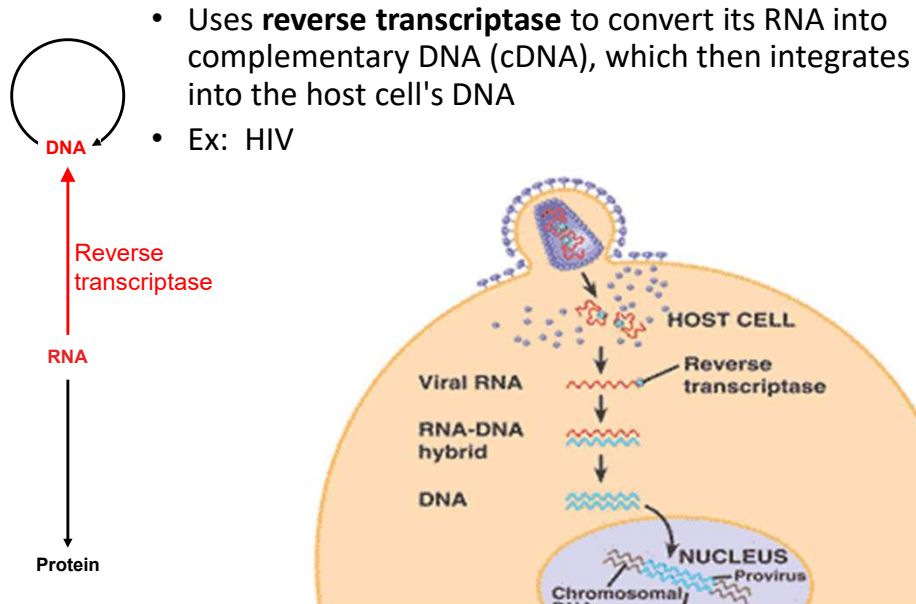
RNA use as templates for both translation and genome replication



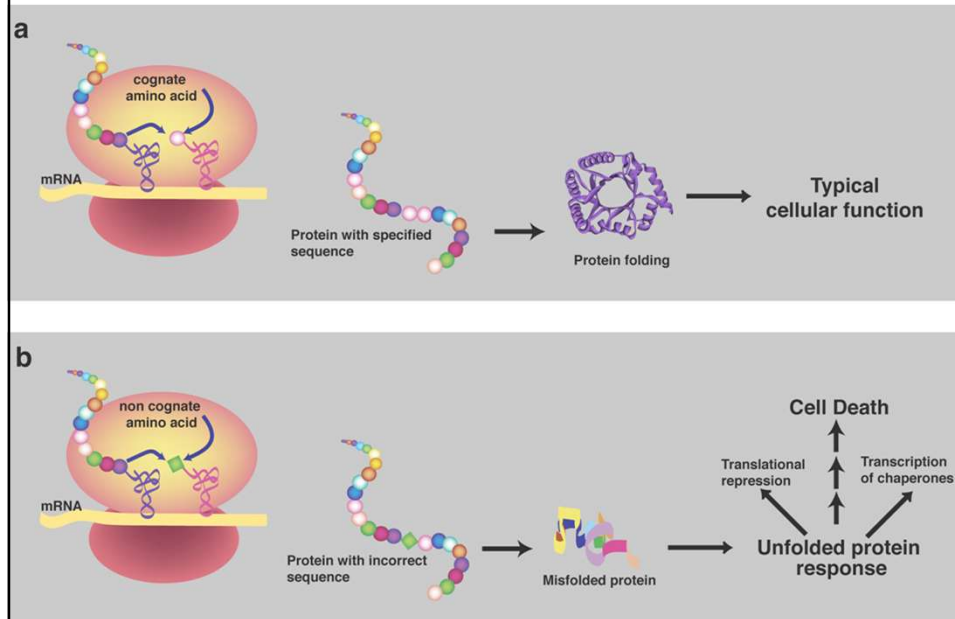
SARS-CoV-2 is a single-stranded RNA (+ssRNA) virus



Retrovirus and Reverse Transcription

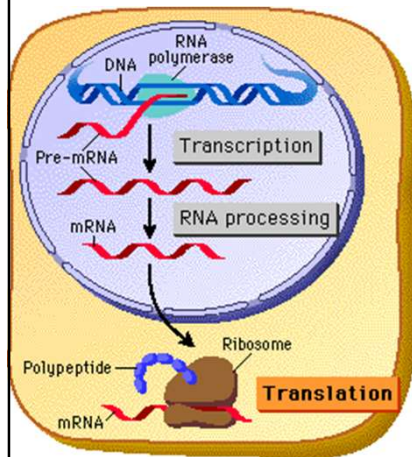


Basic protein science



Residue-by-residue information transfer

Codon: The sequence of 3 nucleotides in DNA/RNA that encodes for a specific amino acid.



(DNA) 5'ATGAGTAACGCG3'
3'TACTCATTGCGC5' template

↓ transcription

(RNA) 5'AUGAGUAACGCG3'
codon

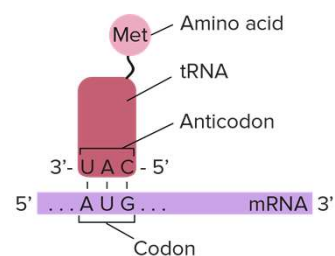
↓ translation

(protein) N-Met-Ser-Asn-Ala-C

Codon

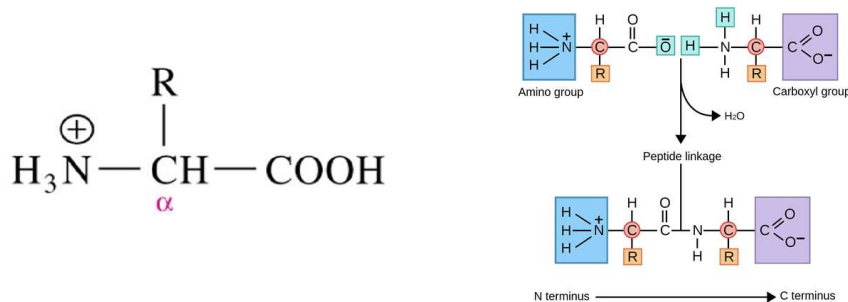
- Three nucleotides on RNA to determine one amino acid
- AUG serves as the "START" codon carrying Met.
- UUA, UGA, and UAG serve as the "STOP" codon.

		2nd base in codon					
		U	C	A	G		
1st base in codon	U	Phe Phe Leu Leu	Ser Ser Ser Ser	Tyr Tyr STOP STOP	Cys Cys STOP Trp	U C A G	3rd base in codon
	C	Leu Leu Leu Leu	Pro Pro Pro Pro	His His Gln Gln	Arg Arg Arg Arg	U C A G	
	A	Ile Ile Ile Met	Thr Thr Thr Thr	Asn Asn Lys Lys	Ser Ser Arg Arg	U C A G	
	G	Val Val Val Val	Ala Ala Ala Ala	Asp Asp Glu Glu	Gly Gly Gly Gly	U C A G	



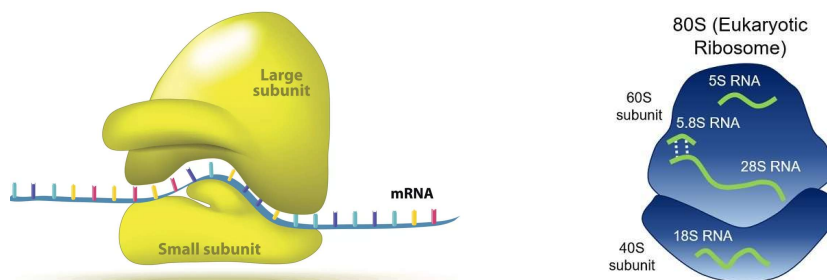
Amino Acids

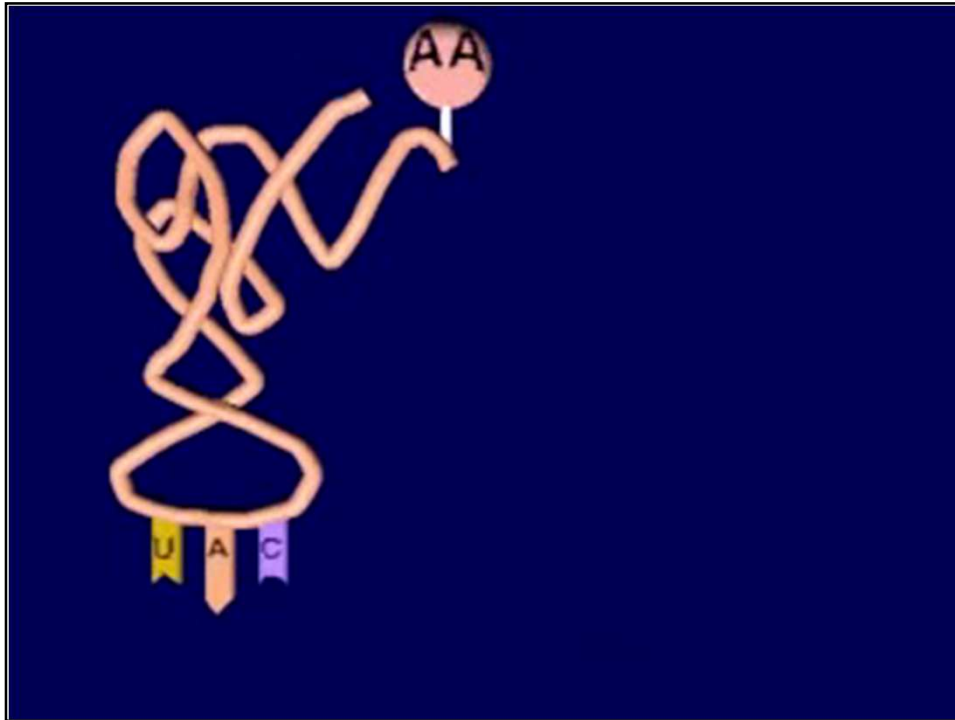
- The basic building block of proteins
- The core structure consists of a central carbon atom, an amino group (NH₃), a carboxyl group (COOH), a hydrogen atom (H), and a side chain (R) with 20 different types.
- C-terminal: The end of the polypeptide with a free COOH.
- N-terminal: The beginning of the polypeptide with a free NH₃.
- Protein is synthesized and read from N-terminal to C-terminal.



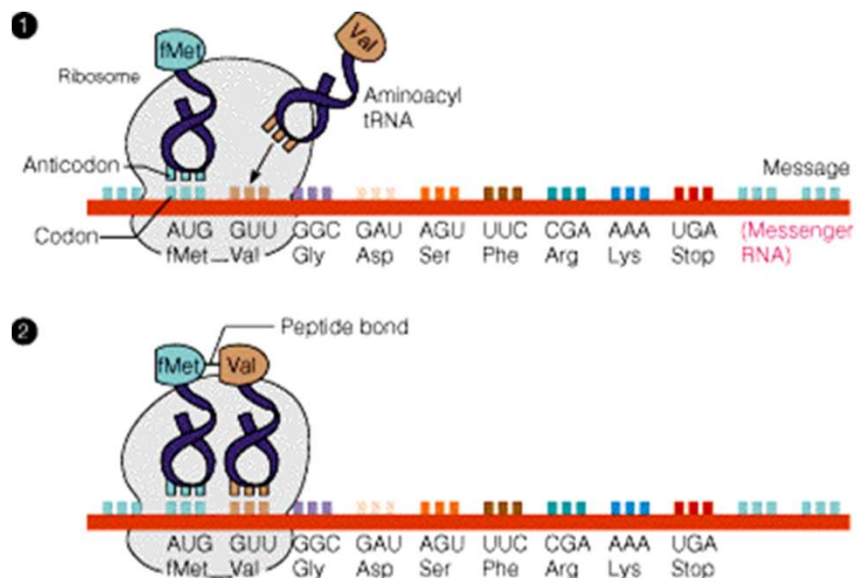
Ribosome - protein synthesizer

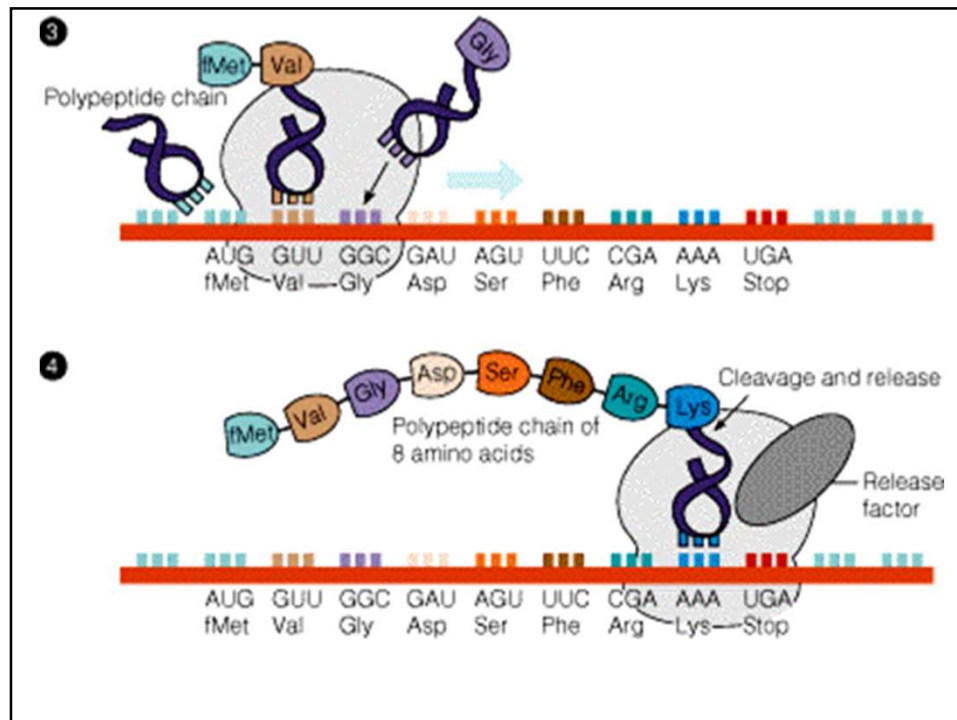
- Protein synthesis takes place in the ribosome
- The Eukaryotic ribosome is made of a small 40S subunit and a large 60S subunit, both containing multiple proteins and rRNAs
- Ribosome move along mRNA and catalysis of peptide bond formation.





Translation is the process of "reading" the codons and linking amino acids together through peptide bonds

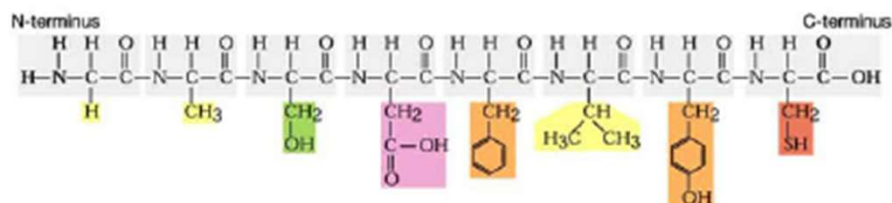




$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_3 \end{array}$ Alanine A	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH} \\ \\ \text{H}_3\text{C}-\text{CH}_3 \end{array}$ Valine V	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{H}_3\text{C}-\text{CH}_2-\text{CH}_3 \end{array}$ Leucine L	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{H}_3\text{C}-\text{CH} \\ \\ \text{CH}_2 \\ \\ \text{CH}_3 \end{array}$ Isoleucine I	$\begin{array}{c} \text{COO}^- \\ \\ \text{HN}-\text{C}-\text{H} \\ \\ 2\text{HC}-\text{CH}_2 \\ \\ \text{CH}_2 \end{array}$ Proline P	Side chains
$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{S} \\ \\ \text{CH}_3 \end{array}$ Methionine M	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_5 \end{array}$ Phenylalanine F	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_8\text{H}_7\text{N} \end{array}$ Tryptophan W	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{H} \end{array}$ Glycine G	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{OH} \end{array}$ Serine S	Hydrophobic (hate water)
$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{HC}-\text{OH} \\ \\ \text{CH}_3 \end{array}$ Threonine T	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{SH} \end{array}$ Cysteine C	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}(=\text{O})\text{NH}_2 \end{array}$ Asparagine N	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}(=\text{O})\text{NH}_2 \end{array}$ Glutamine Q	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_4 \\ \\ \text{OH} \end{array}$ Tyrosine Y	Hydrophilic (like water)
$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}(=\text{O})\text{O}^- \end{array}$ Aspartic Acid D	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}(=\text{O})\text{O}^- \end{array}$ Glutamic Acid E	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH}_3^+ \end{array}$ Lysine K	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH} \\ \\ \text{C}(\text{NH}_2)_2 \end{array}$ Arginine R	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{HC}=\text{C} \\ \quad \backslash \\ \text{NH} \quad \text{NH} \\ \quad \backslash \\ \text{H} \quad \text{H} \end{array}$ Histidine H	negatively and positively charged

Protein vs. peptide vs. polypeptide

- Peptides (from the Greek πεπτίδια, "small digestibles") are short polymers of amino acids (<50 amino acids).
- Polypeptides are >50 amino acid chains **without** folding.
- Proteins are polypeptides **with** particular 3D structures and functions.

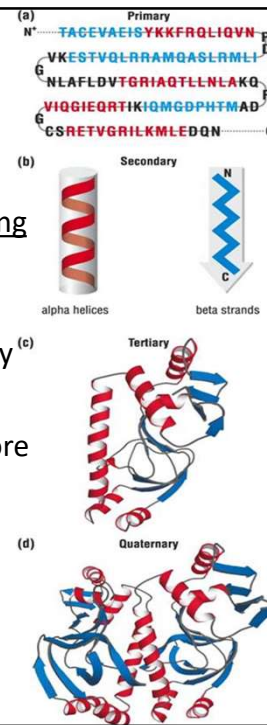
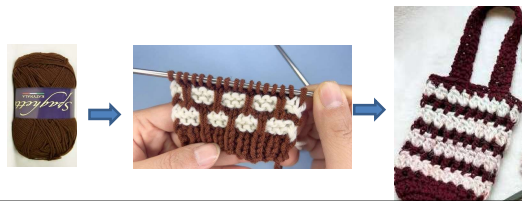


Changes of polypeptide chain after translation

- The peptide chain folds into a specific structure
- Transport to their functional location
- Some amino acids might be removed
- Addition of some modifiers (phosphate, methyl, sugar, lipid, ion, etc.) or interacting with other proteins
- Degradation

Levels of protein structure

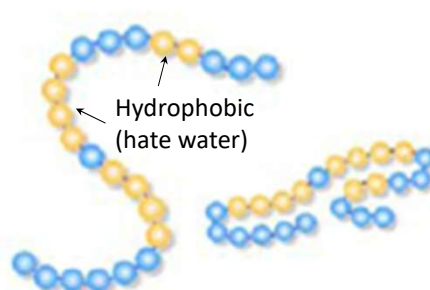
- **Primary structure** - amino acid linear sequence
- **Secondary structure** - regions of regularly repeating conformations of the peptide chain. Two major types are **α -helices** and **β -sheets**.
- **Tertiary structure** - describes the shape of the fully folded polypeptide chain
- **Quaternary structure** - arrangement of two or more proteins into a multi-subunit molecule



The difference between unfolded and folded protein

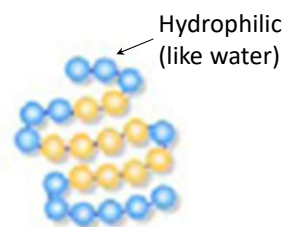
Unfolded protein

Aggregation-prone
Protease-sensitive
Non functional



Properly folded protein

Stable and soluble
Protease-resistant
Functional



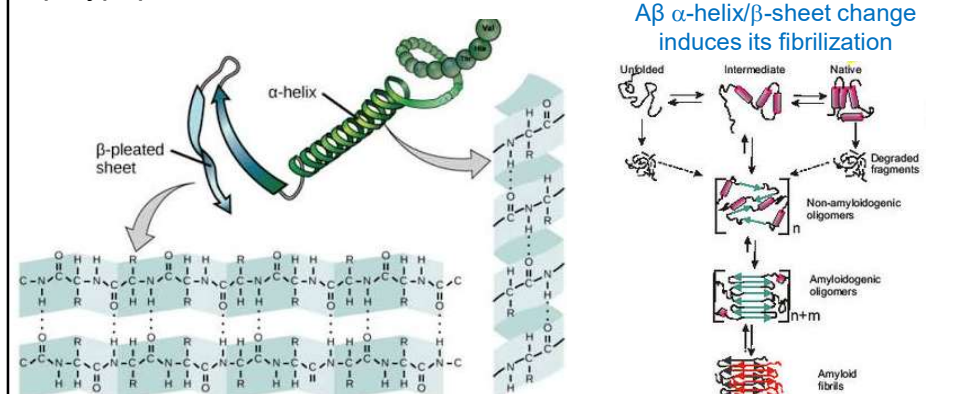
Secondary structure

Strands of polypeptides have distinct characteristic local structural conformations.

α -Helix, the hydrophobic residues hide within the helix

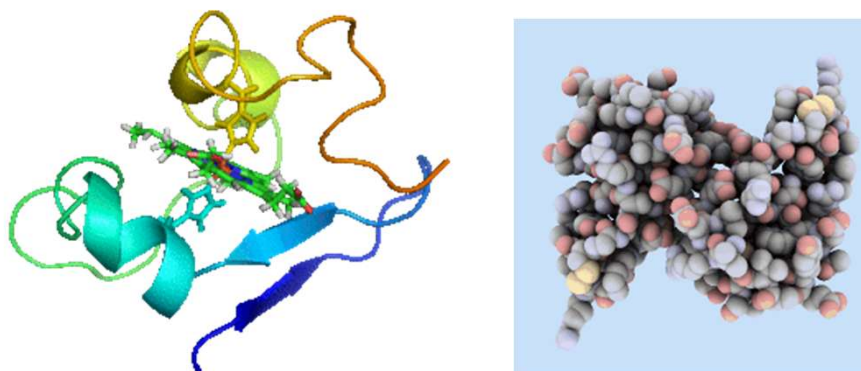
β -Sheet, the hydrophobic residues bury between sheets.

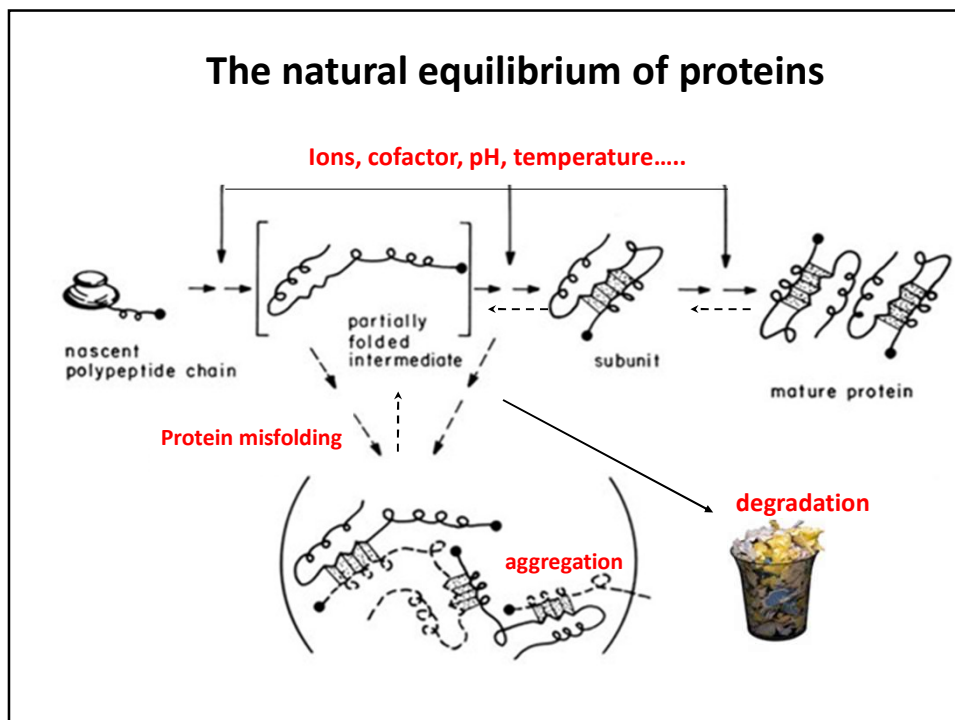
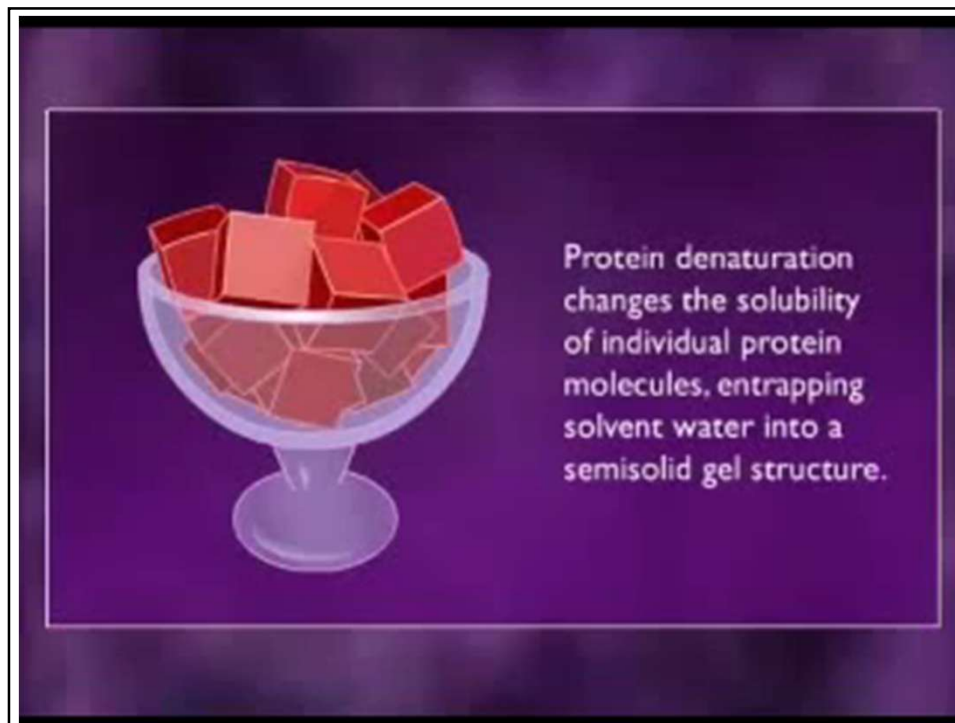
Turns & Loops: Short flexible connector regions allow the polypeptide chain to reverse direction



Tertiary structure

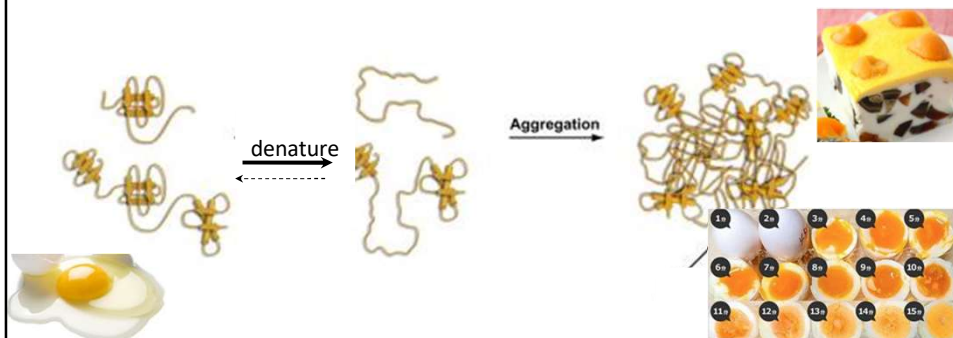
- The overall 3D shape of an polypeptide
- The protein usually bend and twist to achieve maximum stability or lowest energy state.
- The environment are significant determinants of its shape



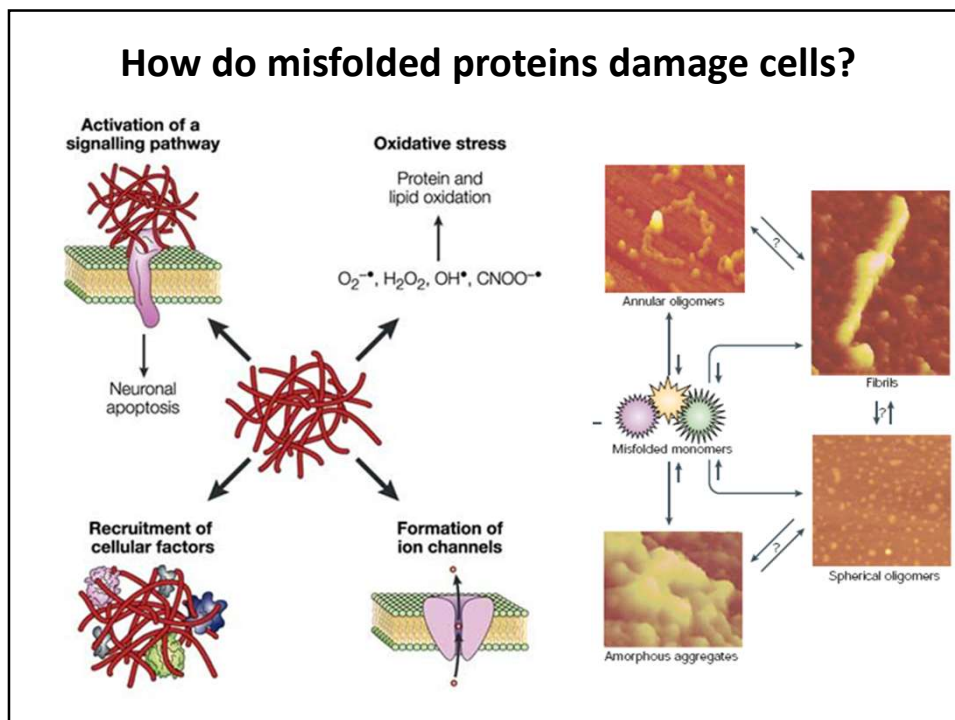


Denaturation and aggregation

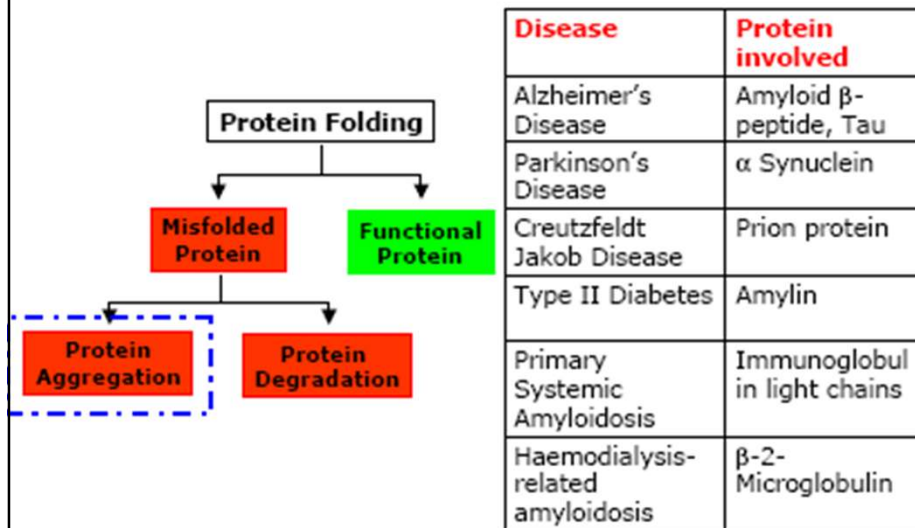
- Transformation of a folded protein to an unfolded state under non-physiological conditions.
 - Usually lost their biological function
 - Denaturing agents include: temperature, pressure, shear, pH, organic solvents, detergents, high salts
 - Denatured proteins have a greater tendency to aggregate



How do misfolded proteins damage cells?

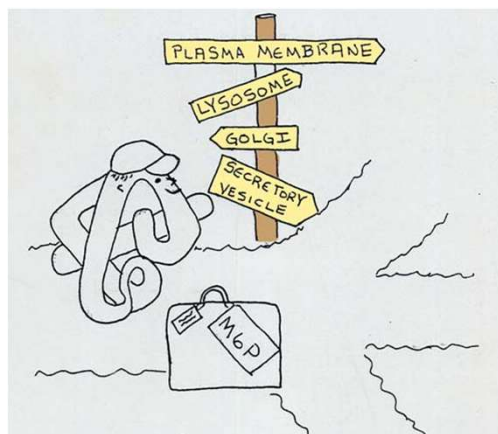


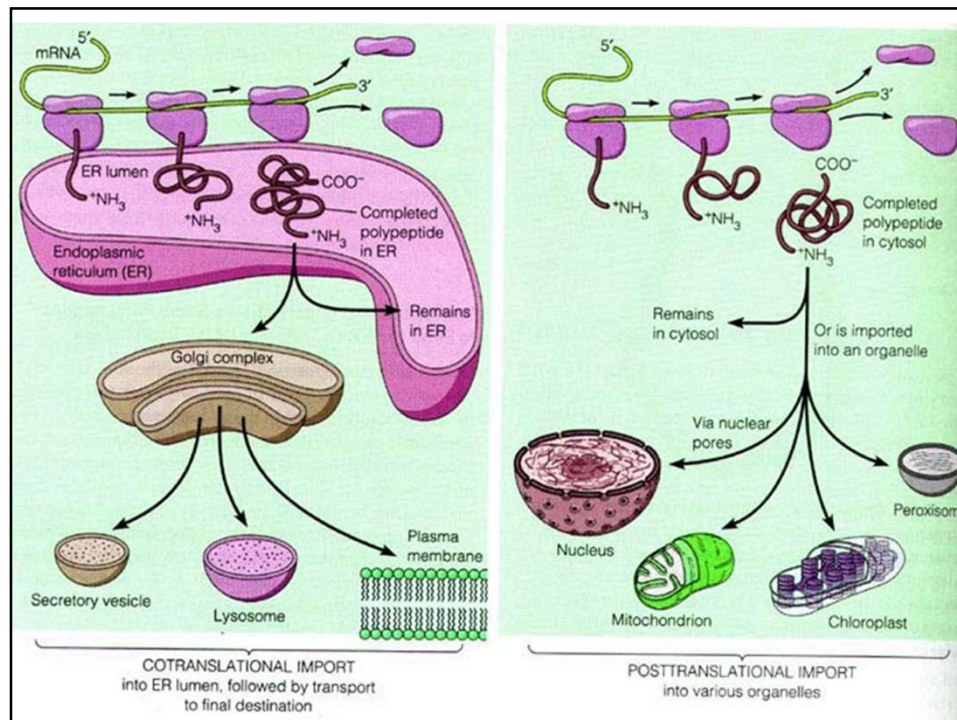
Diseases due to accumulation of misfolded protein



Protein sorting (targeting)

- Transports proteins to the appropriate positions inside or outside of the cell.
- This delivery process is carried out based on information contained in the protein itself.





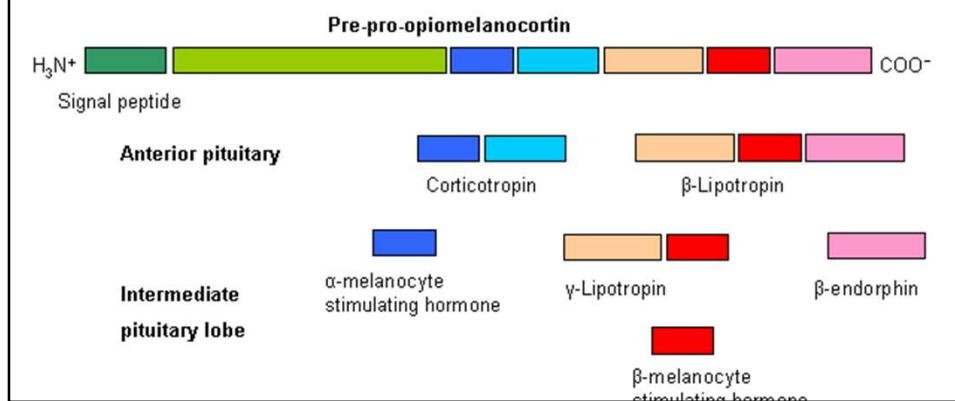
Sequences to control sorting

- A **signal peptide** is a short (3-60 amino acids) peptide chain that directs the transport of a protein to a specific cellular compartment.
- Also called targeting sequences or localization signals.
- Usually removed after targeting

FUNCTION OF SIGNAL SEQUENCE	EXAMPLE OF SIGNAL SEQUENCE
Import into nucleus	-Pro-Pro-Lys-Lys-Lys-Arg-Lys-Val-
Export from nucleus	-Leu-Ala-Leu-Lys-Leu-Ala-Gly-Leu-Asp-Ile-
Import into mitochondria	*H ₃ N-Met-Leu-Ser-Leu-Arg-Gln-Ser-Ile-Arg-Phe-Phe-Lys-Pro-Ala-Thr-Arg-Thr-Leu-Cys-Ser-Ser-Arg-Tyr-Leu-Leu-
Import into plastid	*H ₃ N-Met-Val-Ala-Met-Ala-Met-Ala-Ser-Leu-Gln-Ser-Ser-Met-Ser-Ser-Leu-Ser-Leu-Ser-Ser-Asn-Ser-Phe-Leu-Gly-Gln-Pro-Leu-Ser-Pro-Ile-Thr-Leu-Ser-Pro-Phe-Leu-Gln-Gly-
Import into peroxisomes	-Ser-Lys-Leu-COO ⁻
Import into ER	*H ₃ N-Met-Met-Ser-Phe-Val-Ser-Leu-Leu-Leu-Val-Gly-Ile-Leu-Phe-Trp-Ala-Thr-Glu-Ala-Glu-Gln-Leu-Thr-Lys-Cys-Glu-Val-Phe-Gln-
Return to ER	-Lys-Asp-Glu-Leu-COO ⁻

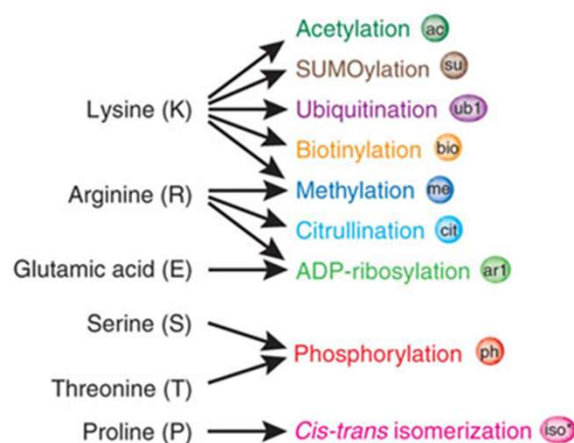
Proteolytic processing

- Cut by **protease**
- the regulated, specific cleavage of peptide bonds in a protein
- Make cell-specific proteolytic products from one polypeptide
- Control protein maturation or activation



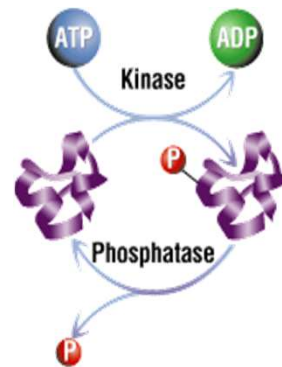
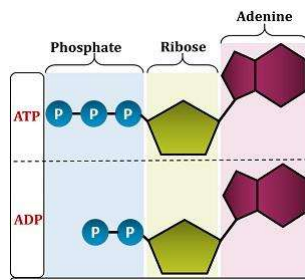
Types of Post-translational modifications

- Proteolytic cleavage
- Glycosylation
- Methylation
- Hydroxylation
- Phosphorylation
- Sulfation
- Acylation
- Carboxylation
- Prenylation
- Selenation
- Formylation
- Disulfide bond formation



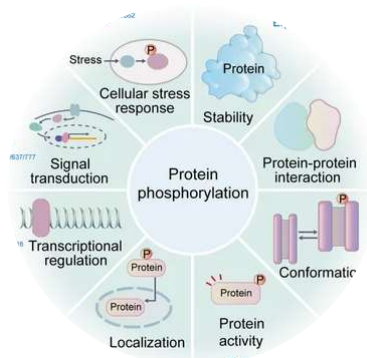
Phosphorylation/Dephosphorylation

- Phosphorylation is the addition of a phosphate (PO_4) group to a protein; Dephosphorylation takes PO_4 out.
- Phosphorylation catalyzed by protein **kinases**; Dephosphorylation by **phosphatases**
- Adenosine triphosphate (ATP) is the phosphate and energy donor, turning into adenosine diphosphate (ADP)



Functions of Phosphorylation

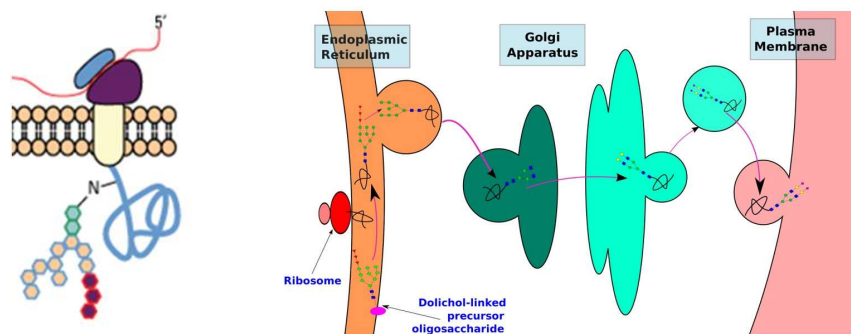
- A fast molecular switch in response to environmental changes.
- It changes the protein's shape to control the following:
 - **Signal Transduction:** passes signals from one protein to another, letting cells react quickly to outside changes.
 - **Enzyme Regulation:** can either turn an enzyme on or off.
 - **Protein Interaction:** regulates protein binding to other proteins/DNA.
 - **Cellular Localization:** moving proteins within cell



- In a eukaryotic cell, the time for a 50 kDa protein to be made (transcription+ translation), folding, PTM, and transportation takes ~10 minutes
- Protein phosphorylation takes < 1 min

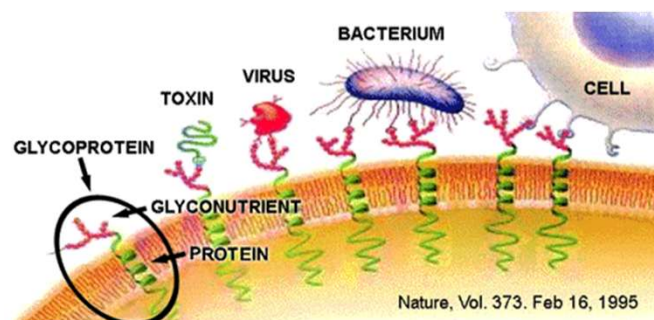
Glycosylation

- Covalently attaching carbohydrate chains (glycans) to target proteins
- Mostly occurring in the endoplasmic reticulum (ER) and Golgi apparatus



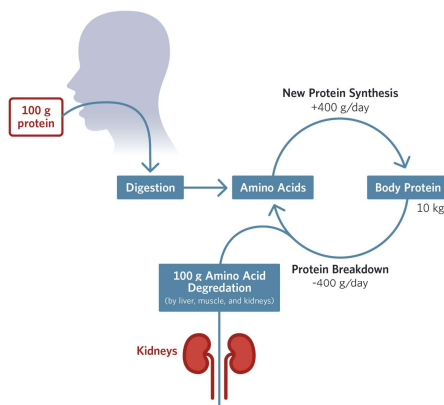
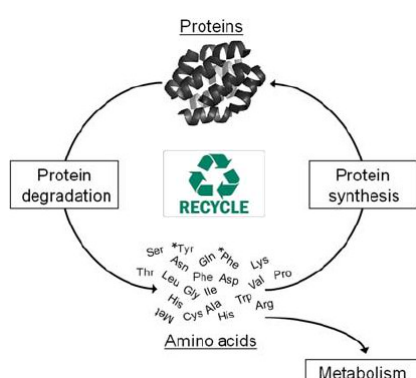
Functions of glycosylation

- Many cell surface and extracellular proteins are glycosylated.
- Stabilize proteins against proteolysis
- Modulation of protein folding
- Provide sorting signals
- Contributes to cell surface molecule recognition

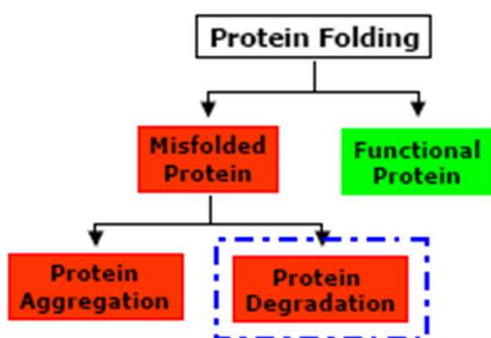


Protein Turnover

- The continuous process of synthesizing new proteins and degrading old or damaged proteins in the body.
- Humans turn over 300-400 g of protein daily, which is higher than the typical daily dietary intake (<100 g)
- Two major pathways: **lysosome and proteasome**



Diseases due to the loss of protein



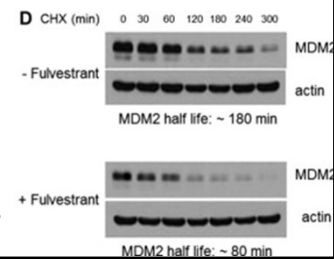
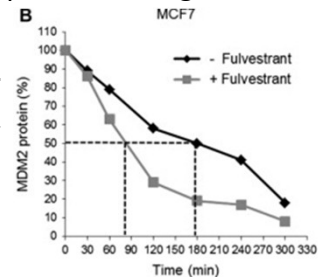
Premature stop codon
 Accelerated breakdown of protein
 Reduced amount of functional protein
 Protein misfolding and premature degradation

Disease	Protein involved
Cystic fibrosis	CFTR
Retinitis Pigmentosa 3	Rhodopsin
α -1 antitrypsin deficiency	α -1 anti trypsin
Cancer	p53
Fabry Disease	α -galactosidase A
Gaucher's disease	β -glucocerebro-sidase
Osteogenesis imperfecta	Type 1 procollagen pro a
Marfan syndrome	Fibrillin

Proteins have variable life-spans

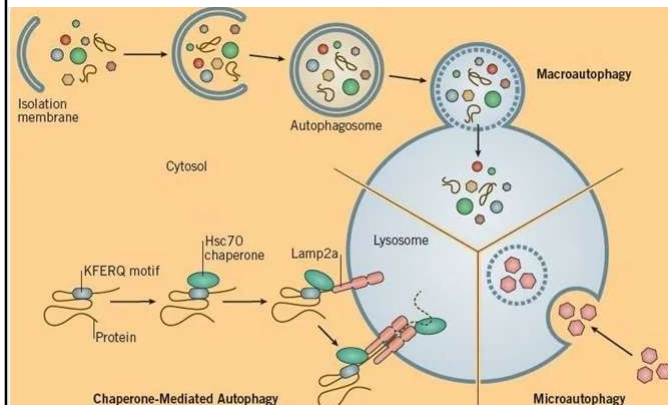
Half-life - Average time for **50%** of the protein pool to be degraded

	Half-life (Hours)
Ornithine decarboxylase	0.2
RNA polymerase I	1.3
Tyrosine aminotransferase	2.0
Serine dehydratase	4.0
PEPcarboxylase	5.0
Aldolase	118
GAPDH	130
cytochrome c	150

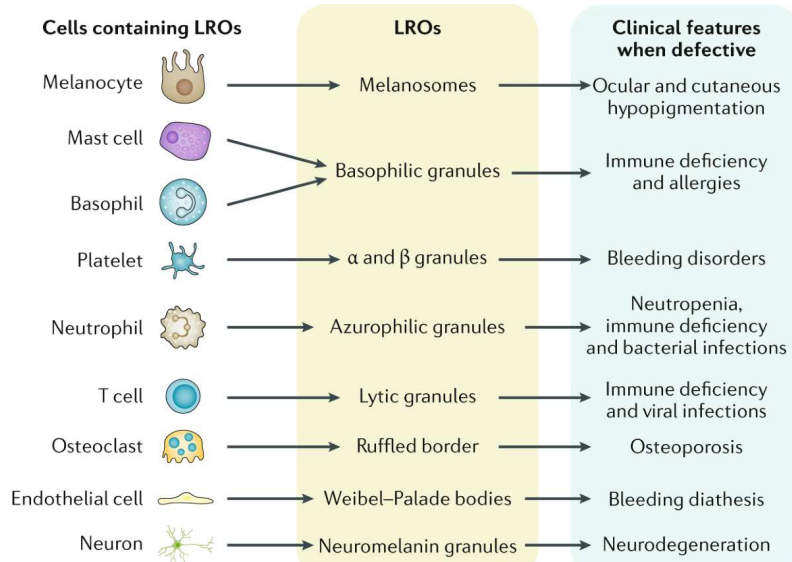


Autophagy-Lysosome Pathway

- A cellular recycling system clears damaged cell parts and proteins.
- Lysosome contain over 50 different proteases.
- **Macroautophagy**: wastes are trapped by an autophagosome that then fuse with a lysosome to digest the contents.
- **Microautophagy**: Lysosomes swallow proteins in the cytoplasm.
- **Chaperone-mediated autophagy**: proteins are marked by helper molecules and pumped directly into the lysosome for breakdown

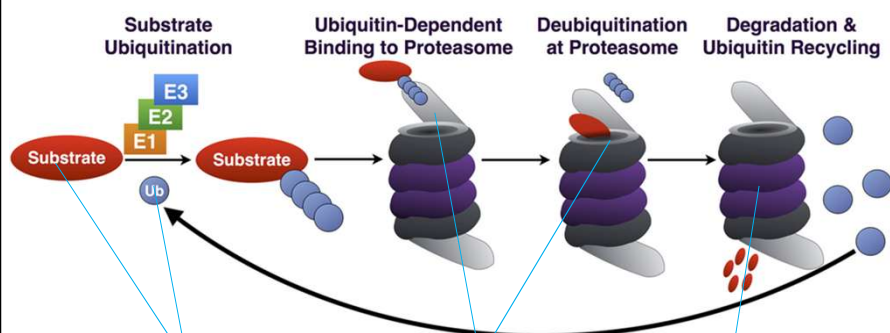


Lysosomal storage disease



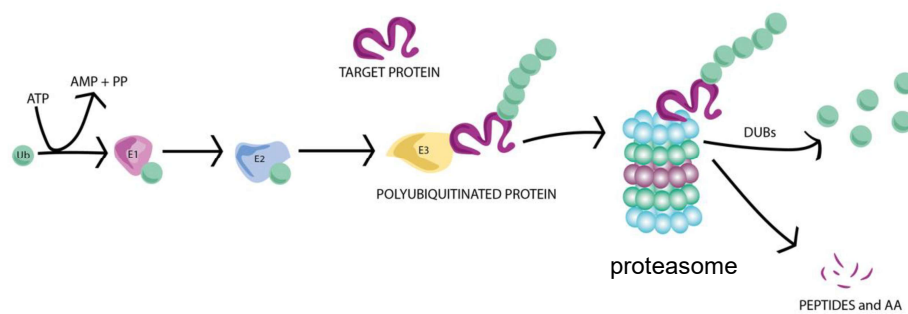
Ubiquitin-proteasome pathway

System to tag and degrade unneeded or damaged proteins.

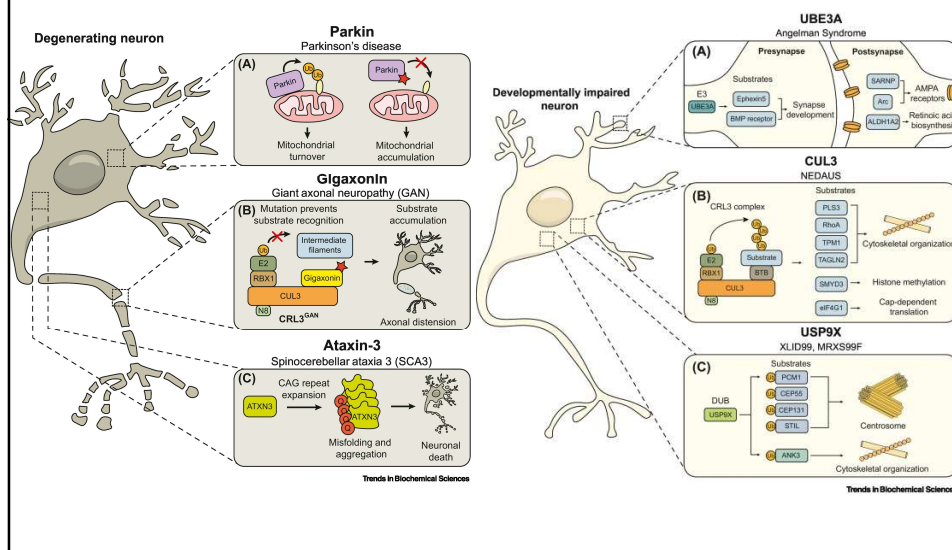


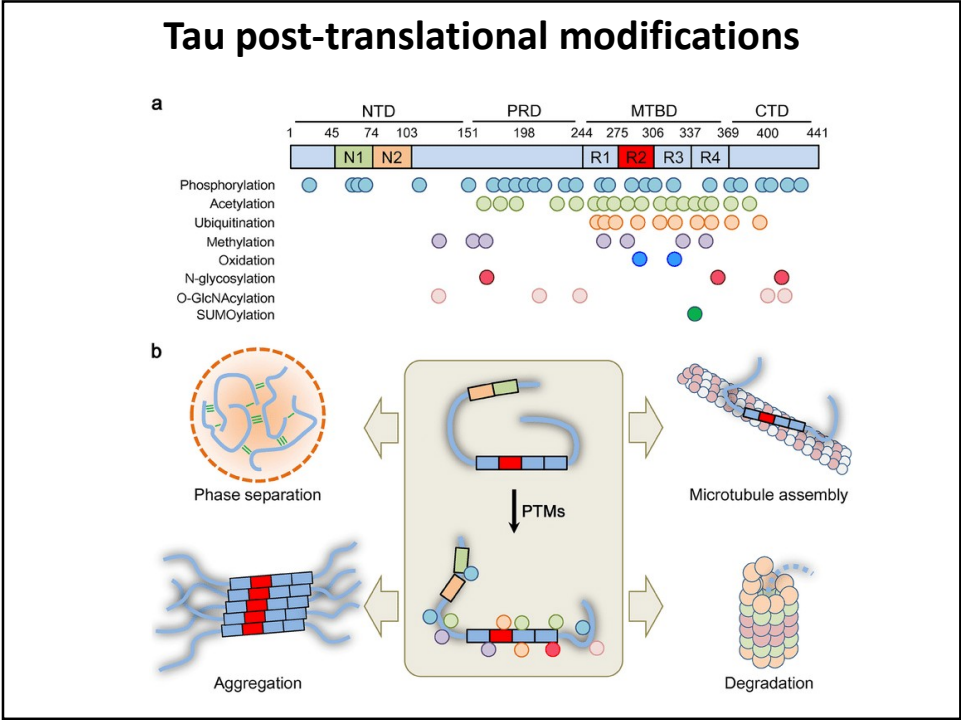
Ubiquitin-proteasome pathway

- Ubiquitin : a highly conserved 76 amino acid protein
- Proteins are targeted for degradation by covalent ligation to polyubiquitin chain with E1/E2/E3 enzymes (require energy)
- Polyubiquitinated proteins are degraded by proteasome
- Proteasome is a large protein complexes containing several proteases inside all eukaryotes



Ubiquitin system mutations in neurological diseases





Tau post-translational modifications

A

Projection domain (acidic region)	Proline-rich domain	Assembly domain Microtubule-binding domain
N 45 N1 N2 103	151 197	244 274 R1 R2 R3 R4 336 368 C 441

B

	FTLD-Tau spectrum (Primary tauopathies)				Secondary tauopathies
Tau Aggregation Characteristics	PID Neuronal cytoplasmic inclusions (Pick bodies) and ramified astrocytic inclusions	PSP Globose tangles in neurons and tufted astrocytic inclusions	CBD Neuronal cytoplasmic inclusions and astrocytic plaques	FTDP-17 Filamentous tau inclusions in neurons and astrocytes	AD Intracellular neurofibrillary tangles composed of tau and Aβ extracellular plaques
Tau Isoform	3R tauopathy	4R tauopathy		3R:4R tauopathy	
Genetic Characteristics	Sporadic (≈80% cases) Familial (≈20% cases)				Sporadic (90% cases) Familial (10% cases)
PTMs Detected	Modification Phosphorylation Acetylation Ubiquitination Methylation SUMOylation Glycation O-GlcNAcylation N-glycosylation Proteolysis	Targeted residues Ser, Thr, and Tyr Lys Lys Lys and Arg Lys Lys Ser and Thr Asn Protease-specific	Diseases observed AD and FTLD-tau AD and FTLD-tau AD and FTLD-tau AD AD AD Only found in healthy brains AD AD and FTLD-tau		